Sahaj Bhadja

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Summary

Software Engineer with 2+ years building object-oriented services in **C#/.NET**, Java, and Python on **AWS** and Azure, with PostgreSQL and SQL Server, background job processing, **troubleshooting**, and reliable production operations.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Database Systems, Analysis of Algorithms, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Object-Oriented Programming, Software Engineering, Operating Systems, Data Retrieval

Experience

USC Dornsife **September 2025 – Present**
Software Engineer *Los Angeles, CA*

- Built fault-tolerant **background services** with FastAPI workers, explicit job-state transitions, and failure isolation, processing 1K+ asynchronous jobs at 3x throughput without cascading failures under concurrent production load.
- Built integration pipelines across **PostgreSQL**, SQL Server, and cloud storage with reconciliation checks and CI/CD releases at 99%+ reliability, and tuned indexes to cut dashboard latency from 1.2s to 350ms.

Grad8 **June 2025 – August 2025**
Software Engineer Intern *Los Angeles, CA*

- Built web services and a React application on **AWS** with MySQL on RDS for upload, rubric, and grading workflows, serving 150+ instructors and 10K+ daily requests in production.
- **Troubleshot** OCR timeouts and model failures in production by adding fallback routing and multithreaded batch processing for 1K+ files per run, keeping APIs stable and cutting feedback turnaround 60%.

Aasma Technology Solutions **August 2023 – July 2024**
Software Developer Engineer – SecureSync *India*

- Built **object-oriented** Java and Python class libraries with Singleton and Factory patterns for cross-platform microservices, covered by 20+ Mockito and PyTest suites in nightly Jenkins pipelines across platforms.
- Redesigned ETL on **AWS Glue** and S3 that transformed JSON, CSV, and Parquet data, boosting pipeline efficiency 68%, and found root causes in API integrations to improve response times 30%.

Aasma Technology Solutions **May 2023 – August 2023**
Full Stack Developer Intern – VibrantChem *India*

- Built **C# .NET** microservices on Azure SQL for authentication, resource allocation, billing, and notifications, defining service boundaries across 8 services and syncing 2M+ records across locations in real time.
- Diagnosed production issues with Sentry, Azure Monitor, and Kibana and shipped fixes through Jenkins CI/CD with Docker packaging and staged releases, cutting deployment time 50% and keeping **reliability** high.

LST Capitals **November 2022 – January 2023**
Web Developer Intern *India*

- Built Spring Boot and Flask APIs for bookings and payments with Firebase **NoSQL** storage and transactional locking to prevent double-booked slots, keeping responses under 200ms across 50K+ records.
- Processed **event-driven** payment webhooks idempotently for refunds and disputes, cutting manual handling about 70%, and built hourly Spark analytics over booking and payment events for internal dashboards.

Skills

Languages: C#, Java, Python, SQL, JavaScript, TypeScript, Bash
Backend: .NET, Spring Boot, FastAPI, Flask, REST APIs, Microservices, Background Jobs, Webhooks
Data: PostgreSQL, SQL Server, MySQL, Azure SQL, NoSQL, MongoDB, Firebase, ETL Pipelines
Cloud: AWS (S3, RDS, EC2, Glue), Azure, Docker, Kubernetes, CI/CD, Jenkins, Linux, Postman
Practices: Object-Oriented Design, Design Patterns, Troubleshooting, Root Cause Analysis, Sentry, Mockito, PyTest, Git, Agile

Projects

Domain Registration Service | *C#, .NET, AWS Lambda, API Gateway, Aurora PostgreSQL, SQS*

- Designed a **.NET** service that handles domain availability checks, registrations, and renewals through API Gateway and Lambda, stores records in **Aurora PostgreSQL**, and queues billing events through SQS.

Domain Lifecycle Workflow | *AWS Step Functions, Lambda, DynamoDB, CloudWatch, Python*

- Designed a **Step Functions** workflow that moves expiring domains through grace, redemption, and deletion states, tracks status in DynamoDB, and raises CloudWatch alarms for **on-call** engineers to follow up.

Chat2SQL Code | *Java, TypeScript, PostgreSQL, SQL Server, MySQL, Docker*

- Extended a Java and TypeScript database workspace supporting 10+ engines including **PostgreSQL**, SQL Server, and MySQL, with schema synchronization and AI-generated SQL queries reaching 82% accuracy across environments.